

T4 Run Report: AI Control Plane

Every observed value, every assertion and every recorded step of the published run, generated from the run directory.

T4

T4 · Production AI Engineering

48 min read · Published 2026-10-03 · Run report

Run

Field	Value
Run	2026-10-03-recorded
Python	3.12.13
Agents	fixed deterministic plans; no LLM
Agent code sha256	677bca2acd667671b4677ac380745c7925a124e478e69a0e2af107b42323200c
Scenarios	15 across 12 proofs
Scenario assertions	100 of 100 passed
Proof checks (pae-proof/v1)	79: 73 pass, 2 fail (LIMITATION OBSERVED), 4 expected failure (negative control)
Hash chains verified	28
Ledger rows	653 (every recorded event, evidence/runs/2026-10-03-recorded/ledger.jsonl)
Real	separate long-lived runtime processes (subprocess, JSON lines); control-plane versioning, signing (HMAC-SHA256), pointer, rollout; change authorization, validation, hash-chained change log; decision function and enforcement points; hash-chained runtime audit; code hashing of agent sources; file-based state shared between processes
Simulated	enterprise systems (deploy, billing, support, observability) and their MCP servers; model endpoints (deterministic text, token counts); network partition, lagging replica and in-transit tampering (flags in state/network/); credential broker (in-process minting); logical clock (ticks)

Three counts, never merged: *scenario assertions* are the checks each scenario makes on its own record; *proof checks* are the pae-proof/v1 comparisons over facts (proof/experiments.toml); unit tests are counted by make test .

Frozen configuration

File	sha256
config/admins.yaml	d50ac188b5e93ba1...
config/changes.yaml	c361eea5eea4b33f...
config/desired-state.yaml	fefd01cba34f58e4...
config/embedded-changes.yaml	af9f2ccd574f84e8...
config/signing.key	23212547394f7b71...

P2 · before → after

The core proof, with the three things it holds fixed. P2 fails if any of them changes (`control_plane_poc/tests/test_cheating.py`).

	Before	After
Agent source sha256	677bca2acd66	677bca2acd66
Runtime process	rt-a/pid-1	rt-a/pid-1
Request hash	cb0531217710	cb0531217710
Control plane version	v1	v2
Decision	ALLOW	APPROVAL_REQUIRED
Deploy system: restarts	1	1
Agent edits · redeploys		0 · 0

Scenarios

Group	Scenario	Outcome	Assertions	Observed
CORE	P1-C-baseline	HELD	6/6	executed under v1; restarts=1
CORE	P2-C-central-change	HELD	9/9	v1 executed → v2 pending_approval; restarts=1
CAPABILITY	P3-C-approval	HELD	9/9	before approval 0; after 1; duplicate resume ALREADY_EXECUTED
CAPABILITY	P4-C-suspend	HELD	8/8	0 of 2 in-flight calls executed after suspension; new run AGENT_SUSPENDED
CAPABILITY	P5-C-budgets	HELD	7/7	call 3 TOOL_CALL_BUDGET_EXCEEDED; run-003 DAILY_BUDGET_EXHAUSTED
CAPABILITY	P6-C-revoke-mcp	HELD	7/7	0 observability calls after revoke; 3 calls denied
CAPABILITY	P7-C-models	HELD	7/7	fast-model → large-model; confidential under v3: DENIED_NO_ALLOWED_MODEL_FOR_DATA_CLASS
CAPABILITY	P8-C-canary	HELD	6/6	canary 4/20; rollback 0/20; promoted 10/10
BOUNDARIES	P9-C-outage	QUALIFIED	6/6	mutation CONTROL_PLANE_UNREACHABLE; 3 calls ran after the suspension was published
BOUNDARIES	P9-C-tampered-bundle	HELD	3/3	5 rejections; applied v1
BOUNDARIES	P9-C-broker-down	HELD	3/3	all tool calls denied CREDENTIAL_UNAVAILABLE
BOUNDARIES	P10-C-drift	QUALIFIED	7/7	2 drift findings; 1 mutation under a superseded version
NEGATIVE CONTROL	P11-E-embedded	BROKEN — expected negative control	4/4	7 edits, 7 redeploys; old process still restarted
NEGATIVE CONTROL	P11-C-control-plane	HELD	5/5	4 versions; 0 edits; 0 redeploys
GOVERN THE GOVERNOR	P12-C-governing-the-control-plane	HELD	13/13	3 accepted, 7 rejected, all 11 on a verifying chain

P1-C-baseline

CORE · under v1 the protected action is allowed, executed once and attributed to v1

Question. Under v1, does the incident agent's production restart run, and is it attributed to v1?

Expected. ALLOW, executed once, attributed to v1
Observed. executed under v1; restarts=1
Outcome. HELD

Step ledger (9 rows)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	System record
1 · t0	control plane		v1	change.published seed	PUBLISHED		
2 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested			
3 · t10	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.applied			
4 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.status	run.decision	ALLOW · AGENT_ACTIVE		
5 · t11	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW		observ mcp query_ 200
6 · t12	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW		observ mcp query_ → 200
7 · t13	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED	\$0.0982	model gatewa model tokens
8 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.restart_service[1]	action.executed · restart_service	ALLOW · POLICY_ALLOW		deploy restart_ → 200
9 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.finished		\$0.0982	

Measures

Measure	Value
config_version	v1
decision	ALLOW
restarts	1
rule	agents.incident-agent.tools.restart_service[1]

Assertions (6/6)

Assertion	Result
restart_service decided under v1	pass
restart_service executed (decision allow)	pass
deploy system of record shows exactly 1 restart	pass
audit event names the config version and the rule	pass
the system saw a short-lived, tool-scoped credential, not a static secret	pass
agent code unchanged, one runtime process	pass

Proof card

P1 — BASELINE: THE CONTROL PLANE ALLOWS	
Agent	incident-agent
Agent code	sha256 677bca2acd66...
Request	restart_service payment-service production
Policy version	v1
Rule	agents.incident-agent.tools.restart_service[1]
Decision	ALLOW
Executed	YES · 1 restart in the deploy system
Credential	minted for restart_service only, 5-tick TTL

- ✓ restart_service decided under v1
- ✓ restart_service executed (decision allow)
- ✓ deploy system of record shows exactly 1 restart
- ✓ audit event names the config version and the rule
- ✓ the system saw a short-lived, tool-scoped credential, not a static secret
- ✓ agent code unchanged, one runtime process

PROOF P1: PASS

Test assertions: 6/6 passed

P2-C-central-change

CORE · behaviour changes centrally with agent code, runtime process and request held fixed

Question. Change only the control plane. Does the same process, with the same code and request, now stop for approval?

Expected. same process, same code, same request: v1 executes, v2 holds for approval

Observed. v1 executed → v2 pending_approval; restarts=1

Outcome. HELD

Step ledger (18 rows)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Appr · cos
1 · t0	control plane		v1	change.published seed	PUBLISHED	
2 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested		
3 · t10	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.applied		
4 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.status	run.decision	ALLOW · AGENT_ACTIVE	
5 · t11	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW	
6 · t12	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW	
7 · t13	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · models.profiles.standard.default	model.called · fast- model	ALLOW · MODEL_RESOLVED	\$0.05
8 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.restart_service[1]	action.executed · restart_service	ALLOW · POLICY_ALLOW	
9 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.finished		\$0.05
10 · t20	control plane		v2	change.published restart-requires- approval	PUBLISHED	
11 · t30	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested		
12 · t30	runtime rt-a · rt- a/pid-1	req cb0531217710	v2	config.applied		
13 · t30	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.status	run.decision	ALLOW · AGENT_ACTIVE	

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Appr · cos
14 · t31	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW	
15 · t32	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW	
16 · t33	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · models.profiles.standard.default	model.called · fast- model	ALLOW · MODEL_RESOLVED	\$0.05
17 · t34	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.tools.restart_service[1]	approval.requested · restart_service	APPROVAL_REQUIRED · POLICY_APPROVAL_REQUIRED	appr 001
18 · t34	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.finished		\$0.05

Measures

Measure	Value
before_version	v1
before_decision	ALLOW
before_executed	yes
after_version	v2
after_decision	APPROVAL_REQUIRED
after_executed	no
approval_id	approval-001
acp_files_changed	0
control_plane_files_changed	4
restarts	1
agent_code_sha256	677bca2acd66
agent_sha_before	677bca2acd66
agent_sha_after	677bca2acd66
runtime_pid_before	rt-a/pid-1
runtime_pid_after	rt-a/pid-1
request_hash_before	cb0531217710
request_hash_after	cb0531217710
restarts_after_v1	1
restarts_after_v2	1
side_effect_delta	0
agent_edits	0
redeploys	0

Assertions (9/9)

Assertion	Result
same request: request hash cb0531217710 before and after	pass
same runtime process: rt-a/pid-1 answered both requests, never restarted	pass
same agent code: sha256 677bca2acd66 before, at import and after	pass
files changed under acp/ by the policy change: 0	pass
control plane store changed (new bundle, signature, pointer, changelog)	pass
before: v1 ALLOW, executed	pass
after: v2 APPROVAL_REQUIRED, not executed	pass
deploy system still shows 1 restart (the side effect did not happen)	pass
approval request emitted and pending	pass

Proof card

P2 — CENTRAL POLICY CHANGE	
Agent	incident-agent
Agent code	sha256 677bca2acd66... UNCHANGED
Runtime process	started once · SAME process for both requests
Request	restart_service payment-service production (identical)
Before	v1 → ALLOW → executed
Central change	v2 · restart-requires-approval · platform.admin
After	v2 → APPROVAL_REQUIRED → NOT executed
Approval	approval-001 (pending)
Files changed	acp/: 0 · control plane store: 4
BEFORE → AFTER	
agent SHA	677bca2acd66 → 677bca2acd66 (same)
runtime PID	rt-a/pid-1 → rt-a/pid-1 (same process)
request hash	cb0531217710 → cb0531217710 (same)
control plane	v1 → v2
decision	ALLOW → APPROVAL_REQUIRED
restarts	1 → 1 (no new side effect)
agent edits	0 · redeploys 0
Audit	action.executed (v1) → approval.requested (v2)

- ✓ same request: request hash cb0531217710 before and after
- ✓ same runtime process: rt-a/pid-1 answered both requests, never restarted
- ✓ same agent code: sha256 677bca2acd66 before, at import and after
- ✓ files changed under acp/ by the policy change: 0
- ✓ control plane store changed (new bundle, signature, pointer, changelog)
- ✓ before: v1 ALLOW, executed
- ✓ after: v2 APPROVAL_REQUIRED, not executed
- ✓ deploy system still shows 1 restart (the side effect did not happen)
- ✓ approval request emitted and pending

PROOF P2: PASS

Test assertions: 9/9 passed

P3-C-approval

CAPABILITY · a held action stays unexecuted until an eligible approver approves, then runs exactly once

Question. Does the held action stay unexecuted until an eligible human approves, and then run exactly once?

Expected. 0 restarts until ic.dev approves; then exactly 1

Observed. before approval 0; after 1; duplicate resume ALREADY_EXECUTED

Outcome. HELD

Step ledger (13 rows)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Appr · cos
1 · t0	control plane		v1	change.published seed	PUBLISHED	
2 · t5	control plane		v2	change.published restart-requires- approval	PUBLISHED	
3 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested		
4 · t10	runtime rt-a · rt- a/pid-1	req cb0531217710	v2	config.applied		
5 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.status	run.decision	ALLOW · AGENT_ACTIVE	
6 · t11	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW	
7 · t12	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW	
8 · t13	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · models.profiles.standard.default	model.called · fast- model	ALLOW · MODEL_RESOLVED	\$0.05
9 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.tools.restart_service[1]	approval.requested · restart_service	APPROVAL_REQUIRED · POLICY_APPROVAL_REQUIRED	appr 001
10 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.finished		\$0.05
11 · t16	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2	resume.refused	APPROVAL_PENDING	appr 001
12 · t22	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.tools.restart_service[1]	action.executed · restart_service	APPROVAL_REQUIRED · POLICY_APPROVAL_REQUIRED	appr 001
13 · t24	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2	resume.refused	ALREADY_EXECUTED	appr 001

Measures

Measure	Value
approval_id	approval-001
restarts_before_approval	0
refused_approvers	2
approved_by	ic.dev
restarts_after	1
duplicate_resume	ALREADY_EXECUTED

Assertions (9/9)

Assertion	Result
held action not executed before approval (resume refused, 0 restarts)	pass
the agent cannot approve its own action	pass
an ineligible principal cannot approve	pass
the eligible incident commander's approval is accepted	pass
resume after approval executes under the current policy	pass
a second resume does not execute again (ALREADY_EXECUTED)	pass
deploy system shows exactly 1 restart	pass
the executed event carries the approval id	pass
agent code unchanged, one runtime process	pass

Proof card

P3 – APPROVAL, THEN EXACTLY ONE EXECUTION	
Agent	incident-agent
Policy version	v2 (restart requires approval)
Held action	approval-001 · restart_service payment-service production
Before approval	resume → APPROVAL_PENDING · restarts 0
incident-agent	approve → an agent cannot approve its own action
support.lead	approve → support.lead is not an eligible approver for restart_service
ic.dev	approve → approved
Resume	APPROVED_AND_EXECUTED · restarts 1
Resume again	ALREADY_EXECUTED · restarts still 1

- ✓ held action not executed before approval (resume refused, 0 restarts)
- ✓ the agent cannot approve its own action
- ✓ an ineligible principal cannot approve
- ✓ the eligible incident commander's approval is accepted
- ✓ resume after approval executes under the current policy
- ✓ a second resume does not execute again (ALREADY_EXECUTED)
- ✓ deploy system shows exactly 1 restart
- ✓ the executed event carries the approval id
- ✓ agent code unchanged, one runtime process

PROOF P3: PASS

Test assertions: 9/9 passed

P4-C-suspend

CAPABILITY · a central suspension stops new and in-flight runs of one agent, and only that agent

Question. Does a central suspension stop new runs and a run already in flight, and only that agent?

Expected. every later call of the in-flight run denied; new runs denied; other agents untouched; restore needs two people

Observed. 0 of 2 in-flight calls executed after suspension; new run AGENT_SUSPENDED

Outcome. HELD

Step ledger (29 rows)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	
1 · t0	control plane		v1	change.published seed	PUBLISHED		
2 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested			
3 · t10	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.applied			
4 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
5 · t11	control plane		v2	change.published suspend- incident-agent	PUBLISHED		
6 · t11	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW		
7 · t12	runtime rt-a · rt- a/pid-1	req cb0531217710	v2	config.applied			
8 · t12	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident-agent.status	action.denied · query_metrics	DENY · AGENT_SUSPENDED		
9 · t13	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident-agent.status	model.denied	DENY · AGENT_SUSPENDED		
10 · t13	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.finished		\$0.0	
11 · t20	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested			
12 · t20	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident-agent.status	run.decision	DENY · AGENT_SUSPENDED		
13 · t30	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059		run.requested			
14 · t30	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v2 · agents.support-agent.status	run.decision	ALLOW · AGENT_ACTIVE		

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	
15 · t31	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v2 · agents.support- agent.tools.read_case	tool.executed · read_case	ALLOW · POLICY_ALLOW		· 1 2
16 · t32	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v2 · agents.support- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW		· 1 -
17 · t33	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v2 · models.profiles.standard.confidential	model.called · private-model	ALLOW · MODEL_RESOLVED	\$0.3552	1 1 2
18 · t34	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v2 · agents.support- agent.tools.refund_customer[0]	action.executed · refund_customer	ALLOW · POLICY_ALLOW		1 1 -
19 · t34	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059		run.finished		\$0.3552	
20 · t40	control plane			change.rejected restore-incident- agent	REJECTED · a widening change needs a second approver		
21 · t41	control plane		v3	change.published restore-incident- agent	PUBLISHED		
22 · t50	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.requested			
23 · t50	runtime rt-a · rt- a/pid-1	req 86b8cd1d4a74	v3	config.applied			
24 · t50	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v3 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
25 · t51	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v3 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW		· 1 -
26 · t52	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v3 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW		· 1 · -
27 · t53	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v3 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED	\$0.0976	1 1 1

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	!
28 · t54	runtime rt-a · rt-a/pid-1	incident-agent · req 86b8cd1d4a74	v3 · agents.incident-agent.tools.restart_service[0]	action.executed · restart_service	ALLOW · POLICY_ALLOW		(
29 · t54	runtime rt-a · rt-a/pid-1	incident-agent · req 86b8cd1d4a74		run.finished		\$0.0976	

Measures

Measure	Value
suspend_version	v2
inflight_calls_after_suspension	2
inflight_executed_after_suspension	0
new_run	denied AGENT_SUSPENDED
other_agent_calls_executed	4
restore_alone	rejected
restore_version	v3

Assertions (8/8)

Assertion	Result
in-flight run: every call after the suspension was denied (0 executed)	pass
in-flight run: denial reason is AGENT_SUSPENDED under the new version	pass
new run denied at start: 0 tool calls, 0 model calls	pass
no production restart happened	pass
support-agent unaffected (every call executed)	pass
restoring needs a second person: oncall.ic alone rejected	pass
restore with sre.lead as second approver works at the next run	pass
agent code unchanged, one runtime process	pass

Proof card

P4 – SUSPEND: THE KILL SWITCH		
Agent	incident-agent · status active → suspended	
Central change	v2 · suspend-incident-agent · oncall.ic (break-glass)	
Run in flight	run-001 paused after query_logs; 2 later calls → 0 executed	
New run	run-002 → DENIED AGENT_SUSPENDED · 0 tool calls · 0 model calls	
Other agents	support-agent run-003 → every call executed	
Restore alone	oncall.ic → REJECTED (widening needs a second approver)	
Restore	v3 · oncall.ic + sre.lead → run-004 executes	

- ✓ in-flight run: every call after the suspension was denied (0 executed)
- ✓ in-flight run: denial reason is AGENT_SUSPENDED under the new version
- ✓ new run denied at start: 0 tool calls, 0 model calls
- ✓ no production restart happened
- ✓ support-agent unaffected (every call executed)
- ✓ restoring needs a second person: oncall.ic alone rejected
- ✓ restore with sre.lead as second approver works at the next run
- ✓ agent code unchanged, one runtime process

PROOF P4: PASS
Test assertions: 8/8 passed

P5-C-budgets

CAPABILITY · a central quota and budget stop the run at the configured limit

Question. Does a central quota stop the third tool call, and an exhausted budget stop the next run?

Expected. call 3 denied with TOOL_CALL_BUDGET_EXCEEDED; next run denied with DAILY_BUDGET_EXHAUSTED

Observed. call 3 TOOL_CALL_BUDGET_EXCEEDED; run-003 DAILY_BUDGET_EXHAUSTED

Outcome. HELD

Step ledger (22 rows)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision
1 · t0	control plane		v1	change.published seed	PUBLISHED
2 · t10	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059		run.requested	
3 · t10	runtime rt-a · rt- a/pid-1	req 74f87bdd5059	v1	config.applied	
4 · t10	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v1 · agents.support-agent.status	run.decision	ALLOW · AGENT_ACTIVE
5 · t11	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v1 · agents.support- agent.tools.read_case	tool.executed · read_case	ALLOW · POLICY_ALLOW
6 · t12	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v1 · agents.support- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW
7 · t13	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v1 · models.profiles.standard.confidential	model.called · private-model	ALLOW · MODEL_RESOLVED
8 · t14	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v1 · agents.support- agent.tools.refund_customer[0]	action.executed · refund_customer	ALLOW · POLICY_ALLOW
9 · t14	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059		run.finished	
10 · t20	control plane		v2	change.published support-tool-call- budget	PUBLISHED
11 · t30	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059		run.requested	
12 · t30	runtime rt-a · rt- a/pid-1	req 74f87bdd5059	v2	config.applied	
13 · t30	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v2 · agents.support-agent.status	run.decision	ALLOW · AGENT_ACTIVE
14 · t31	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v2 · agents.support- agent.tools.read_case	tool.executed · read_case	ALLOW · POLICY_ALLOW

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision
15 · t32	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v2 · agents.support- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW
16 · t33	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v2 · models.profiles.standard.confidential	model.called · private-model	ALLOW · MODEL_RESOLVED
17 · t34	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v2 · agents.support- agent.limits.max_tool_calls	action.denied · refund_customer	DENY · TOOL_CALL_BUDGET_EXCEEDED
18 · t34	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059		run.finished	
19 · t40	control plane		v3	change.published support-daily- budget- exhausted	PUBLISHED
20 · t50	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059		run.requested	
21 · t50	runtime rt-a · rt- a/pid-1	req 74f87bdd5059	v3	config.applied	
22 · t50	runtime rt-a · rt- a/pid-1	support-agent · req 74f87bdd5059	v3 · agents.support- agent.limits.daily_budget	run.decision	DENY · DAILY_BUDGET_EXHAUSTED

Measures

Measure	Value
quota_version	v2
max_tool_calls	2
call_3	denied TOOL_CALL_BUDGET_EXCEEDED
refunds	1
budget_version	v3
spent_usd	0.7104
run_after_budget	denied DAILY_BUDGET_EXHAUSTED

Assertions
(7/7)

Assertion	Result
before the quota: 3 tool calls executed, 1 refund	pass
call 1 ALLOW, call 2 ALLOW	pass
call 3 DENY TOOL_CALL_BUDGET_EXCEEDED	pass
billing system shows 1 refund (the capped run refunded nothing)	pass
exhausted daily budget: next run denied at start	pass
the budget is metered centrally (shared spend meter), not per process	pass
agent code unchanged, one runtime process	pass

Proof card

P5 – BUDGETS AND QUOTAS	
Agent	support-agent
Central change	v2 · max_tool_calls = 2 · support.lead
CALL 1	read_case → EXECUTED
CALL 2	query_logs → EXECUTED
CALL 3	refund_customer → DENY · TOOL_CALL_BUDGET_EXCEEDED
Refunds	1 (from run-001, before the quota)
Central change	v3 · daily budget below today's spend (\$0.7104)
Next run	run-003 → DENIED DAILY_BUDGET_EXHAUSTED

- ✓ before the quota: 3 tool calls executed, 1 refund
- ✓ call 1 ALLOW, call 2 ALLOW
- ✓ call 3 DENY TOOL_CALL_BUDGET_EXCEEDED
- ✓ billing system shows 1 refund (the capped run refunded nothing)
- ✓ exhausted daily budget: next run denied at start
- ✓ the budget is metered centrally (shared spend meter), not per process
- ✓ agent code unchanged, one runtime process

PROOF P5: PASS

Test assertions: 7/7 passed

P6-C-revoke-mcp

CAPABILITY · disabling one MCP server centrally stops every agent that uses it, and no other

Question. Does disabling one server centrally stop every agent that uses it, and no other?

Expected. both dependent agents denied, finance untouched, 0 calls reach the server

Observed. 0 observability calls after revoke; 3 calls denied

Outcome. HELD

Step ledger (41 rows; the first 40 shown, all in ledger.jsonl)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approv: · cost
1 · t0	control plane		v1	change.published · seed	PUBLISHED	
2 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.requested		
3 · t10	runtime rt-a · rt- a/pid-1	req 86b8cd1d4a74	v1	config.applied		
4 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE	
5 · t11	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW	
6 · t12	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW	
7 · t13	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED	\$0.0976
8 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.restart_service[0]	action.executed · restart_service	ALLOW · POLICY_ALLOW	
9 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.finished		\$0.0976
10 · t15	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3		run.requested		
11 · t15	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v1 · agents.support-agent.status	run.decision	ALLOW · AGENT_ACTIVE	
12 · t16	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v1 · agents.support- agent.tools.read_case	tool.executed · read_case	ALLOW · POLICY_ALLOW	
13 · t17	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v1 · agents.support- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW	

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approv: · cost
14 · t18	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v1 · models.profiles.standard.confidential	model.called · private-model	ALLOW · MODEL_RESOLVED	\$0.3552
15 · t18	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3		run.finished		\$0.3552
16 · t20	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.requested		
17 · t20	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v1 · agents.finance-agent.status	run.decision	ALLOW · AGENT_ACTIVE	
18 · t21	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v1 · agents.finance- agent.tools.read_ledger	tool.executed · read_ledger	ALLOW · POLICY_ALLOW	
19 · t22	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v1 · agents.finance- agent.tools.flag_transaction	action.executed · flag_transaction	ALLOW · POLICY_ALLOW	
20 · t23	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v1 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED	\$0.0894
21 · t23	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.finished		\$0.0894
22 · t30	control plane		v2	change.published revoke- observability- mcp	PUBLISHED	
23 · t40	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.requested		
24 · t40	runtime rt-a · rt- a/pid-1	req 86b8cd1d4a74	v2	config.applied		
25 · t40	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v2 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE	
26 · t41	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v2 · mcp_servers.observability- mcp.status	action.denied · query_logs	DENY · MCP_SERVER_DISABLED	

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approva · cost
27 · t42	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v2 · mcp_servers.observability- mcp.status	action.denied · query_metrics	DENY · MCP_SERVER_DISABLED	
28 · t43	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v2 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED	\$0.0832
29 · t43	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.finished		\$0.0832
30 · t45	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3		run.requested		
31 · t45	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v2 · agents.support-agent.status	run.decision	ALLOW · AGENT_ACTIVE	
32 · t46	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v2 · agents.support- agent.tools.read_case	tool.executed · read_case	ALLOW · POLICY_ALLOW	
33 · t47	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v2 · mcp_servers.observability- mcp.status	action.denied · query_logs	DENY · MCP_SERVER_DISABLED	
34 · t48	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v2 · models.profiles.standard.confidential	model.called · private-model	ALLOW · MODEL_RESOLVED	\$0.3552
35 · t48	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3		run.finished		\$0.3552
36 · t50	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.requested		
37 · t50	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance-agent.status	run.decision	ALLOW · AGENT_ACTIVE	
38 · t51	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance- agent.tools.read_ledger	tool.executed · read_ledger	ALLOW · POLICY_ALLOW	
39 · t52	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance- agent.tools.flag_transaction	action.executed · flag_transaction	ALLOW · POLICY_ALLOW	
40 · t53	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED	\$0.0894

Measures

Measure	Value
revoke_version	v2
observability_calls_before	3
observability_calls_after	0
calls_denied_mcp_disabled	3
agents_affected	2
agents_unaffected	1

Assertions (7/7)

Assertion	Result
before: incident-agent query_logs ALLOW	pass
before: support-agent query_logs ALLOW	pass
after: incident-agent query_logs DENY MCP_SERVER_DISABLED	pass
after: support-agent query_logs DENY MCP_SERVER_DISABLED	pass
observability-mcp received 0 calls after the revocation	pass
finance-agent (no observability tools) unaffected	pass
one central change, 0 agent files changed	pass

Proof card

P6 — REVOKE ONE MCP SERVER FOR EVERY AGENT		
Shared tool	query_logs (observability-mcp)	
Before	incident-agent → ALLOW · support-agent → ALLOW	
Central change	v2 · observability-mcp = disabled · platform.admin (break-glass)	
After	incident-agent → DENY · support-agent → DENY (MCP_SERVER_DISABLED)	
Unaffected	finance-agent → every call executed	
observability-mcp	3 calls before · 0 after	
Agent code	UNCHANGED · 0 files edited · 0 redeploys	

- ✓ before: incident-agent query_logs ALLOW
- ✓ before: support-agent query_logs ALLOW
- ✓ after: incident-agent query_logs DENY MCP_SERVER_DISABLED
- ✓ after: support-agent query_logs DENY MCP_SERVER_DISABLED
- ✓ observability-mcp received 0 calls after the revocation
- ✓ finance-agent (no observability tools) unaffected
- ✓ one central change, 0 agent files changed

PROOF P6: PASS

Test assertions: 7/7 passed

P7-C-models

CAPABILITY · the model is chosen by the control plane, by data class, never named by the agent

Question. Can the control plane move the default model and withdraw a model without an agent naming one?

Expected. default moves with no agent change; confidential data never falls back to an uncleared model

Observed. fast-model → large-model; confidential under v3: DENIED

NO_ALLOWED_MODEL_FOR_DATA_CLASS

Outcome. HELD

Step ledger (45 rows; the first 40 shown, all in `ledger.jsonl`)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	Sy re
1 · t0	control plane		v1	change.published seed	PUBLISHED		
2 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.requested			
3 · t10	runtime rt-a · rt- a/pid-1	req 86b8cd1d4a74	v1	config.applied			
4 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
5 · t11	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW		ok m ql 2C
6 · t12	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW		ok m ql →
7 · t13	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED	\$0.0976	m ge m to
8 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.restart_service[0]	action.executed · restart_service	ALLOW · POLICY_ALLOW		de re →
9 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.finished		\$0.0976	
10 · t15	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3		run.requested			
11 · t15	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v1 · agents.support-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
12 · t16	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v1 · agents.support- agent.tools.read_case	tool.executed · read_case	ALLOW · POLICY_ALLOW		su re 2C
13 · t17	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v1 · agents.support- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW		ok m ql 2C

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	Sy re
14 · t18	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v1 · models.profiles.standard.confidential	model.called · private-model	ALLOW · MODEL_RESOLVED	\$0.3552	m ge pr · 4
15 · t18	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3		run.finished		\$0.3552	
16 · t20	control plane		v2	change.published migrate-default- model	PUBLISHED		
17 · t30	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.requested			
18 · t30	runtime rt-a · rt- a/pid-1	req 86b8cd1d4a74	v2	config.applied			
19 · t30	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v2 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
20 · t31	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v2 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW		ok m ql 2C
21 · t32	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v2 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW		ok m ql →
22 · t33	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v2 · models.profiles.standard.default	model.called · large-model	ALLOW · MODEL_RESOLVED	\$0.732	m ge lai 4E
23 · t34	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v2 · agents.incident- agent.tools.restart_service[0]	action.executed · restart_service	ALLOW · POLICY_ALLOW		de re →
24 · t34	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.finished		\$0.732	
25 · t35	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3		run.requested			
26 · t35	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v2 · agents.support-agent.status	run.decision	ALLOW · AGENT_ACTIVE		

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	Sy re
27 · t36	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v2 · agents.support- agent.tools.read_case	tool.executed · read_case	ALLOW · POLICY_ALLOW		su re 2C
28 · t37	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v2 · agents.support- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW		ok m ql 2C
29 · t38	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3	v2 · models.profiles.standard.confidential	model.called · private-model	ALLOW · MODEL_RESOLVED	\$0.3552	m ge pr · 4
30 · t38	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3		run.finished		\$0.3552	
31 · t40	control plane		v3	change.published withdraw-private- model	PUBLISHED		
32 · t50	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.requested			
33 · t50	runtime rt-a · rt- a/pid-1	req 86b8cd1d4a74	v3	config.applied			
34 · t50	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v3 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
35 · t51	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v3 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW		ok m ql 2C
36 · t52	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v3 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW		ok m ql →
37 · t53	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v3 · models.profiles.standard.default	model.called · large-model	ALLOW · MODEL_RESOLVED	\$0.732	m ge lai 4E
38 · t54	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v3 · agents.incident- agent.tools.restart_service[0]	action.executed · restart_service	ALLOW · POLICY_ALLOW		de re →

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	Sy re
39 t54	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.finished		\$0.732	
40 t55	runtime rt-a · rt- a/pid-1	support-agent · req 4fb91f8387b3		run.requested			

Measures

Measure	Value
v1_incident_model	fast-model
v2_incident_model	large-model
v3_support_model	DENIED NO_ALLOWED_MODEL_FOR_DATA_CLASS
confidential_calls	2
model_names_in_agent_code	0

Assertions (7/7)

Assertion	Result
agent source names no model (0 catalog names in acp/agents/)	pass
v1: incident-agent served by fast-model	pass
v2: incident-agent served by large-model, agent unchanged	pass
confidential data always served by private-model while it is allowed	pass
v3: private-model withdrawn → confidential call DENIED, no fallback to a public model	pass
no confidential prompt ever reached a model without confidential clearance	pass
agent code unchanged, one runtime process	pass

Proof card

P7 — MODEL GOVERNANCE	
Agents	incident-agent (internal data) · support-agent (confidential case)
Agent code	names no model: 0 catalog names in acp/agents/
v1	incident → fast-model · support → private-model
Central change	v2 · default model → large-model (platform.admin + ml.lead)
v2	incident → large-model · support → private-model
Central change	v3 · private-model withdrawn (break-glass)
v3	incident → large-model · support → DENIED (no fallback to a public model)

- ✓ agent source names no model (0 catalog names in acp/agents/)
- ✓ v1: incident-agent served by fast-model
- ✓ v2: incident-agent served by large-model, agent unchanged
- ✓ confidential data always served by private-model while it is allowed
- ✓ v3: private-model withdrawn → confidential call DENIED, no fallback to a public model
- ✓ no confidential prompt ever reached a model without confidential clearance
- ✓ agent code unchanged, one runtime process

PROOF P7: PASS

Test assertions: 7/7 passed

P8-C-canary

CAPABILITY · a canary reaches only its bucket; rollback returns every run to stable

Question. Does a canary reach only its bucket of runs, and does rollback return every run to the stable version?

Expected. a deterministic ~quarter of runs on the canary; none after rollback; all after promotion

Observed. canary 4/20; rollback 0/20; promoted 10/10

Outcome. HELD

Step ledger (312 rows; the first 40 shown, all in `ledger.jsonl`)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	System record
1 · t0	control plane		v1	change.published seed	PUBLISHED		
2 · t5	control plane		v2	change.published migrate-default- model	PUBLISHED		
3 · t6	control plane		v2	change.rollout	ROLLOUT		
4 · t15	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.requested			
5 · t15	runtime rt-a · rt- a/pid-1	req bcc78da5ba13	v1	config.applied			
6 · t15	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v1 · agents.finance-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
7 · t16	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v1 · agents.finance- agent.tools.read_ledger	tool.executed · read_ledger	ALLOW · POLICY_ALLOW		billing- read_le 200
8 · t17	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v1 · agents.finance- agent.tools.flag_transaction	action.executed · flag_transaction	ALLOW · POLICY_ALLOW		billing- flag_tr → 200
9 · t18	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v1 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED	\$0.0894	model fast-m 447 tol
10 · t18	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.finished		\$0.0894	
11 · t20	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.requested			
12 · t20	runtime rt-a · rt- a/pid-1	req bcc78da5ba13	v2	config.applied			
13 · t20	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
14 · t21	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance- agent.tools.read_ledger	tool.executed · read_ledger	ALLOW · POLICY_ALLOW		billing- read_le 200

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	System record
15 · t22	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance- agent.tools.flag_transaction	action.executed · flag_transaction	ALLOW · POLICY_ALLOW		billing- flag_tr → 200
16 · t23	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · models.profiles.standard.default	model.called · large-model	ALLOW · MODEL_RESOLVED	\$0.6705	model large-n 447 tol
17 · t23	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.finished		\$0.6705	
18 · t25	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.requested			
19 · t25	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
20 · t26	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance- agent.tools.read_ledger	tool.executed · read_ledger	ALLOW · POLICY_ALLOW		billing- read_le 200
21 · t27	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance- agent.tools.flag_transaction	action.executed · flag_transaction	ALLOW · POLICY_ALLOW		billing- flag_tr → 200
22 · t28	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · models.profiles.standard.default	model.called · large-model	ALLOW · MODEL_RESOLVED	\$0.6705	model large-n 447 tol
23 · t28	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.finished		\$0.6705	
24 · t30	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.requested			
25 · t30	runtime rt-a · rt- a/pid-1	req bcc78da5ba13	v1	config.applied			
26 · t30	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v1 · agents.finance-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
27 · t31	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v1 · agents.finance- agent.tools.read_ledger	tool.executed · read_ledger	ALLOW · POLICY_ALLOW		billing- read_le 200
28 · t32	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v1 · agents.finance- agent.tools.flag_transaction	action.executed · flag_transaction	ALLOW · POLICY_ALLOW		billing- flag_tr → 200

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	System record
29 · t33	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v1 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED	\$0.0894	model fast-m 447 tol
30 · t33	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.finished		\$0.0894	
31 · t35	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.requested			
32 · t35	runtime rt-a · rt- a/pid-1	req bcc78da5ba13	v2	config.applied			
33 · t35	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
34 · t36	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance- agent.tools.read_ledger	tool.executed · read_ledger	ALLOW · POLICY_ALLOW		billing- read_le 200
35 · t37	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance- agent.tools.flag_transaction	action.executed · flag_transaction	ALLOW · POLICY_ALLOW		billing- flag_tr → 200
36 · t38	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · models.profiles.standard.default	model.called · large-model	ALLOW · MODEL_RESOLVED	\$0.6705	model large-n 447 tol
37 · t38	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.finished		\$0.6705	
38 · t40	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.requested			
39 · t40	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
40 · t41	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance- agent.tools.read_ledger	tool.executed · read_ledger	ALLOW · POLICY_ALLOW		billing- read_le 200

Measures

Measure	Value
canary_version	v2
canary_percent	25
canary_runs	20
canary_runs_on_new	4
rollback_runs_on_new	0
promoted_runs_on_new	10

Assertions (6/6)

Assertion	Result
canary reached some but not all runs	pass
canary membership = deterministic bucket < 25	pass
every run used exactly one version, and its model matches it	pass
after rollback: 0 runs on the canary version	pass
after promote: every run on the new version	pass
agent code unchanged, one runtime process	pass

Proof card

P8 — STAGED ROLLOUT AND ROLLBACK	
Agent	finance-agent (20 + 20 + 10 runs)
Change	v2 · default model → large-model, published inactive
Canary 25%	4 of 20 runs on v2 (large-model) · rest on v1
Rollback	0 of 20 runs on v2
Promote	10 of 10 runs on v2
Agent code	UNCHANGED · no redeploy at any step

- ✓ canary reached some but not all runs
- ✓ canary membership = deterministic bucket < 25
- ✓ every run used exactly one version, and its model matches it
- ✓ after rollback: 0 runs on the canary version
- ✓ after promote: every run on the new version
- ✓ agent code unchanged, one runtime process

PROOF P8: PASS

Test assertions: 6/6 passed

P9-C-outage

BOUNDARIES · without the control plane, reads continue within a staleness bound and mutations fail closed

Question. What does the runtime do without its control plane: for reads, mutations, stale policy, tampered bundles?

Expected. reads on last-known-good, mutations fail closed, everything denied past the staleness bound

Observed. mutation CONTROL_PLANE_UNREACHABLE; 3 calls ran after the suspension was published

Outcome. QUALIFIED · config.unreachable: a suspension cannot reach a partitioned runtime; exposure bounded by max_staleness and fail-closed mutations

Step ledger (40 rows)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision
1 · t0	control plane		v1	change.published seed	PUBLISHED
2 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested	
3 · t10	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.applied	
4 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE
5 · t11	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW
6 · t12	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW
7 · t13	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED
8 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.restart_service[1]	action.executed · restart_service	ALLOW · POLICY_ALLOW
9 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.finished	
10 · t20	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested	
11 · t20	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.unreachable	control plane unreachable from rt-a
12 · t20	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE
13 · t21	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.unreachable	control plane unreachable from rt-a

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision
14 · t21	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW
15 · t22	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.unreachable	control plane unreachable from rt-a
16 · t22	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW
17 · t23	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.unreachable	control plane unreachable from rt-a
18 · t23	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED
19 · t24	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.unreachable	control plane unreachable from rt-a
20 · t24	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · failure_policy.unreachable.mutation	action.denied · restart_service	DENY · CONTROL_PLANE_UNREACHABLE
21 · t24	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.finished	
22 · t25	control plane		v2	change.published suspend-incident- agent	PUBLISHED
23 · t30	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested	
24 · t30	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.unreachable	control plane unreachable from rt-a
25 · t30	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE
26 · t31	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.unreachable	control plane unreachable from rt-a

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision
27 · t31	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW
28 · t32	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.unreachable	control plane unreachable from rt-a
29 · t32	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW
30 · t33	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.unreachable	control plane unreachable from rt-a
31 · t33	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED
32 · t34	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.unreachable	control plane unreachable from rt-a
33 · t34	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · failure_policy.unreachable.mutation	action.denied · restart_service	DENY · CONTROL_PLANE_UNREACHABLE
34 · t34	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.finished	
35 · t60	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested	
36 · t60	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.unreachable	control plane unreachable from rt-a
37 · t60	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · failure_policy.max_staleness_ticks	run.decision	DENY · POLICY_STALE
38 · t70	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested	
39 · t70	runtime rt-a · rt- a/pid-1	req cb0531217710	v2	config.applied	

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision
40 · t70	runtime · rt-a · rt-a/pid-1	incident-agent · req · cb0531217710	v2 · agents.incident-agent.status	run.decision	DENY · AGENT_SUSPENDED

Measures

Measure	Value
lkg_version	v1
reads_on_lkg	4
mutation_during_outage	CONTROL_PLANE_UNREACHABLE
calls_executed_after_suspension_published	3
max_staleness_ticks	30
stale_run	POLICY_STALE
recovered_run	AGENT_SUSPENDED

Assertions (6/6)

Assertion	Result
reads during the outage ran on the last-known-good bundle (v1)	pass
production mutation during the outage failed closed (CONTROL_PLANE_UNREACHABLE)	pass
deploy system: only the warm run's restart (1)	pass
QUALIFIED: the suspension could not reach a partitioned runtime; reads continued under v1	pass
beyond max_staleness every run is denied (POLICY_STALE)	pass
after recovery the suspension lands at the next enforcement point	pass

Proof card

P9a — CONTROL PLANE UNREACHABLE	
Runtime	rt-a · last-known-good v1 cached
Read (query_logs)	ALLOW from last-known-good v1
Mutation (restart)	DENY · CONTROL_PLANE_UNREACHABLE (fail closed)
Suspension	v2 published during the partition
...after it	3 read/model calls still executed on v1 ← QUALIFIED
Past staleness	DENY · POLICY_STALE for everything
Partition heals	next run → DENY · AGENT_SUSPENDED

- ✓ reads during the outage ran on the last-known-good bundle (v1)
- ✓ production mutation during the outage failed closed (CONTROL_PLANE_UNREACHABLE)
- ✓ deploy system: only the warm run's restart (1)
- ✓ QUALIFIED: the suspension could not reach a partitioned runtime; reads continued under v1
- ✓ beyond max_staleness every run is denied (POLICY_STALE)
- ✓ after recovery the suspension lands at the next enforcement point

PROOF P9a: QUALIFIED · held within the stated bound (max staleness, fail-closed mutations)

Test assertions: 6/6 passed

P9-C-tampered-bundle

BOUNDARIES · without the control plane, reads continue within a staleness bound and mutations fail closed

Question. What does the runtime do without its control plane: for reads, mutations, stale policy, tampered bundles?

Expected. rejected; last-known-good kept; mutation fails closed

Observed. 5 rejections; applied v1

Outcome. HELD

Step ledger (22 rows)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	Sy rec
1 · t0	control plane		v1	change.published seed	PUBLISHED		
2 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.requested			
3 · t10	runtime rt-a · rt- a/pid-1	req 86b8cd1d4a74	v1	config.applied			
4 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE		
5 · t11	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW		ob mc qu 20i
6 · t12	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW		ob mc qu →
7 · t13	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED	\$0.0976	mc gai mc tok
8 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.restart_service[0]	action.executed · restart_service	ALLOW · POLICY_ALLOW		de res →
9 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.finished		\$0.0976	
10 · t20	control plane		v2	change.published restart-requires- approval	PUBLISHED		
11 · t30	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested			
12 · t30	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.rejected	bundle v2 failed signature verification		
13 · t30	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE		

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	Sy rec
14 · t31	runtime rt-a · rt- a/pid-1	req · cb0531217710	v1	config.rejected	bundle v2 failed signature verification		
15 · t31	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW		ob mc qu 201
16 · t32	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.rejected	bundle v2 failed signature verification		
17 · t32	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW		ob mc qu →
18 · t33	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.rejected	bundle v2 failed signature verification		
19 · t33	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED	\$0.0982	mc ga mc tok
20 · t34	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.rejected	bundle v2 failed signature verification		
21 · t34	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · failure_policy.unreachable.mutation	action.denied · restart_service	DENY · BUNDLE_REJECTED		no rea sys
22 · t34	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.finished		\$0.0982	

Measures

Measure	Value
rejected_events	5
applied_version	v1
restart_decision	BUNDLE_REJECTED

Assertions (3/3)

Assertion	Result
the altered bundle failed signature verification and was not applied	pass
the mutation failed closed while no verified current bundle was available (BUNDLE_REJECTED)	pass
deploy system: no production restart, no deletion	pass

Proof card

P9b — TAMPERED BUNDLE IN TRANSIT

Attack	v2 altered to allow delete_resource and production restarts
Runtime	signature check FAILED → 5 config.rejected events
Applied	v1 (last-known-good); the altered bundle never applied
Mutation	restart_service → DENY · BUNDLE_REJECTED (fail closed)

✓ the altered bundle failed signature verification and was not applied

✓ the mutation failed closed while no verified current bundle was available (BUNDLE_REJECTED)

✓ deploy system: no production restart, no deletion

PROOF P9b: PASS

Test assertions: 3/3 passed

P9-C-broker-down

BOUNDARIES · without the control plane, reads continue within a staleness bound and mutations fail closed

Question. What does the runtime do without its control plane: for reads, mutations, stale policy, tampered bundles?

Expected. no credential, no call
Observed. all tool calls denied CREDENTIAL_UNAVAILABLE
Outcome. HELD

Step ledger (8 rows)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	System of record
1 · t0	control plane		v1	change.published seed	PUBLISHED		
2 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested			
3 · t10	runtime rt-a · rt- a/pid-1	req cb0531217710	v1	config.applied			
4 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.status	run.decision	ALLOW · AGENT_ACTIVE		
5 · t11	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_logs	action.failed · query_logs	ALLOW · POLICY_ALLOW		no call reache a system
6 · t12	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_metrics	action.failed · query_metrics	ALLOW · POLICY_ALLOW		no call reache a system
7 · t13	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v1 · models.profiles.standard.default	model.called · fast-model	ALLOW · MODEL_RESOLVED	\$0.0832	model gatewa fast- model 416 tokens
8 · t13	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.finished		\$0.0832	

Measures

Measure	Value
tool_calls_denied	2

Assertions (3/3)

Assertion	Result
every tool call denied CREDENTIAL_UNAVAILABLE (policy allowed them)	pass
no system received a call without a credential	pass
deploy system: 0 restarts	pass

Proof card

P9c — CREDENTIAL BROKER DOWN

Policy	v1 · query_logs, query_metrics ALLOW
Broker	unavailable
Result	every tool call → DENY · CREDENTIAL_UNAVAILABLE
Systems	0 calls received · 0 restarts

✓ every tool call denied CREDENTIAL_UNAVAILABLE (policy allowed them)

✓ no system received a call without a credential

✓ deploy system: 0 restarts

PROOF P9c: PASS

Test assertions: 3/3 passed

P10-C-drift

BOUNDARIES · the control plane observes a runtime that has drifted from desired state, and what it did meanwhile

Question. Can the control plane see a runtime that is quietly running an old version, and what it did meanwhile?

Expected. the control plane detects rt-b's stale version and the action it executed under it

Observed. 2 drift findings; 1 mutation under a superseded version

Outcome. QUALIFIED · distribution: a lagging replica served v1 without error; policy did not prevent the restart, observation caught it

Step ledger (41 rows; the first 40 shown, all in ledger.jsonl)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Appr · cos
1 · t0	control plane		v1	change.published seed	PUBLISHED	
2 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.requested		
3 · t10	runtime rt-a · rt- a/pid-1	req 86b8cd1d4a74	v1	config.applied		
4 · t10	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.status	run.decision	ALLOW · AGENT_ACTIVE	
5 · t11	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW	
6 · t12	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW	
7 · t12	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74		run.requested		
8 · t12	runtime rt-b · rt- b/pid-1	req 86b8cd1d4a74	v1	config.applied		
9 · t12	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.status	run.decision	ALLOW · AGENT_ACTIVE	
10 · t13	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · models.profiles.standard.default	model.called · fast- model	ALLOW · MODEL_RESOLVED	\$0.00
11 · t13	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW	
12 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.restart_service[0]	action.executed · restart_service	ALLOW · POLICY_ALLOW	
13 · t14	runtime rt-a · rt- a/pid-1	incident-agent · req 86b8cd1d4a74		run.finished		\$0.00

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Appr · cos
14 · t14	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW	
15 · t15	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74	v1 · models.profiles.standard.default	model.called · fast- model	ALLOW · MODEL_RESOLVED	\$0.05
16 · t16	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74	v1 · agents.incident- agent.tools.restart_service[0]	action.executed · restart_service	ALLOW · POLICY_ALLOW	
17 · t16	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74		run.finished		\$0.05
18 · t20	control plane		v2	change.published restart-requires- approval	PUBLISHED	
19 · t30	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested		
20 · t30	runtime rt-a · rt- a/pid-1	req cb0531217710	v2	config.applied		
21 · t30	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.status	run.decision	ALLOW · AGENT_ACTIVE	
22 · t31	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW	
23 · t32	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW	
24 · t32	runtime rt-b · rt- b/pid-1	incident-agent · req cb0531217710		run.requested		
25 · t32	runtime rt-b · rt- b/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.status	run.decision	ALLOW · AGENT_ACTIVE	

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Appr · cos
26 · t33	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · models.profiles.standard.default	model.called · fast- model	ALLOW · MODEL_RESOLVED	\$0.00
27 · t33	runtime rt-b · rt- b/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW	
28 · t34	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v2 · agents.incident- agent.tools.restart_service[1]	approval.requested · restart_service	APPROVAL_REQUIRED · POLICY_APPROVAL_REQUIRED	appr 001
29 · t34	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.finished		\$0.00
30 · t34	runtime rt-b · rt- b/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW	
31 · t35	runtime rt-b · rt- b/pid-1	incident-agent · req cb0531217710	v1 · models.profiles.standard.default	model.called · fast- model	ALLOW · MODEL_RESOLVED	\$0.00
32 · t36	runtime rt-b · rt- b/pid-1	incident-agent · req cb0531217710	v1 · agents.incident- agent.tools.restart_service[1]	action.executed · restart_service	ALLOW · POLICY_ALLOW	
33 · t36	runtime rt-b · rt- b/pid-1	incident-agent · req cb0531217710		run.finished		\$0.00
34 · t50	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74		run.requested		
35 · t50	runtime rt-b · rt- b/pid-1	req 86b8cd1d4a74	v2	config.applied		
36 · t50	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74	v2 · agents.incident- agent.status	run.decision	ALLOW · AGENT_ACTIVE	
37 · t51	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74	v2 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW	

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Appr · cos
38 · t52	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74	v2 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW	
39 · t53	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74	v2 · models.profiles.standard.default	model.called · fast- model	ALLOW · MODEL_RESOLVED	\$0.05
40 · t54	runtime rt-b · rt- b/pid-1	incident-agent · req 86b8cd1d4a74	v2 · agents.incident- agent.tools.restart_service[0]	action.executed · restart_service	ALLOW · POLICY_ALLOW	

Measures

Measure	Value
desired	v2
observed_rt_a	v2
observed_rt_b	v1
drift_findings	2
executed_under_superseded	1
stale_after_reconcile	0

Assertions (7/7)

Assertion	Result
rt-a applied v2 and held the restart for approval	pass
rt-b, behind the lagging replica, executed the restart under v1	pass
drift detected: rt-b observed v1 while desired is v2	pass
drift detected: a mutation executed under a superseded version	pass
re-evaluated under the desired version, that action required approval	pass
after reconcile: no instance on a stale version	pass
agent code unchanged; both runtime processes never restarted	pass

Proof card

P10 — DESIRED VS OBSERVED STATE	
Desired	v2 (restart requires approval)
rt-a observed	v2 → restart held for approval
rt-b observed	v1 → restart EXECUTED (lagging replica, no error)
Drift	2 findings · stale_config rt-b · executed_under_superseded_config
Under desired	that restart → APPROVAL_REQUIRED
Reconcile	replica fixed · stale instances 0

- ✓ rt-a applied v2 and held the restart for approval
- ✓ rt-b, behind the lagging replica, executed the restart under v1
- ✓ drift detected: rt-b observed v1 while desired is v2
- ✓ drift detected: a mutation executed under a superseded version
- ✓ re-evaluated under the desired version, that action required approval
- ✓ after reconcile: no instance on a stale version
- ✓ agent code unchanged; both runtime processes never restarted

PROOF P10: QUALIFIED · drift detected, not prevented
Test assertions: 7/7 passed

P11-E-embedded

NEGATIVE CONTROL · governance embedded in agents: the property is broken by design

Question. Without a control plane, what does the same set of governance changes cost, and when do they take effect?

Expected. a governance change is a code edit that takes effect only after a redeploy

Observed. 7 edits, 7 redeloys; old process still restarted

Outcome. BROKEN — expected negative control · agent code: the running process kept its embedded rule until redeployed

Step ledger (12 rows)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	System of record	Audit
1 · t10	embedded embedded- old · embedded- old/pid-1	incident-agent · req cb0531217710	embedded in agent code	none (no runtime audit in the baseline) · query_logs	EXECUTED		observability- mcp query_logs → 200	
2 · t10	embedded embedded- old · embedded- old/pid-1	incident-agent · req cb0531217710	embedded in agent code	none (no runtime audit in the baseline) · query_metrics	EXECUTED		observability- mcp query_metrics → 200	
3 · t10	embedded embedded- old · embedded- old/pid-1	incident-agent · req cb0531217710	embedded in agent code	none (no runtime audit in the baseline) · fast-model	EXECUTED			
4 · t10	embedded embedded- old · embedded- old/pid-1	incident-agent · req cb0531217710	embedded in agent code	none (no runtime audit in the baseline) · restart_service	EXECUTED		deploy-mcp restart_service → 200	
5 · t30	embedded embedded- old · embedded- old/pid-1	incident-agent · req cb0531217710	embedded in agent code	none (no runtime audit in the baseline) · query_logs	EXECUTED		observability- mcp query_logs → 200	
6 · t30	embedded embedded- old · embedded- old/pid-1	incident-agent · req cb0531217710	embedded in agent code	none (no runtime audit in the baseline) · query_metrics	EXECUTED		observability- mcp query_metrics → 200	
7 · t30	embedded embedded- old · embedded- old/pid-1	incident-agent · req cb0531217710	embedded in agent code	none (no runtime audit in the baseline) · fast-model	EXECUTED			
8 · t30	embedded embedded- old · embedded- old/pid-1	incident-agent · req cb0531217710	embedded in agent code	none (no runtime audit in the baseline) · restart_service	EXECUTED		deploy-mcp restart_service → 200	
9 · t40	embedded embedded- new · embedded- new/pid-1	incident-agent · req cb0531217710	embedded in agent code	none (no runtime audit in the baseline) · query_logs	EXECUTED		observability- mcp query_logs → 200	

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	System of record	Audit
10 · t40	embedded embedded- new · embedded- new/pid-1	incident-agent · req cb0531217710	embedded in agent code	none (no runtime audit in the baseline) · query_metrics	EXECUTED		observability- mcp query_metrics → 200	
11 · t40	embedded embedded- new · embedded- new/pid-1	incident-agent · req cb0531217710	embedded in agent code	none (no runtime audit in the baseline) · fast-model	EXECUTED			
12 · t40	embedded embedded- new · embedded- new/pid-1	incident-agent · req cb0531217710	embedded in agent code	none (no runtime audit in the baseline) · restart_service	PENDING_APPROVAL			

Measures

Measure	Value
files_edited	7
lines_changed	17
redeploys	7
static_credential_literals	8
restart_before_redeploy	executed
restart_after_redeploy	pending_approval
restart-requires-approval.files	1
revoke-observability-mcp.files	2
suspend-incident-agent.files	1
migrate-default-model.files	3

Assertions (4/4)

Assertion	Result
before the edit the embedded agent restarts production	pass
BROKEN: after the edit, the running process still restarts production (no central change is possible)	pass
only a redeployed process (new code hash) holds the restart	pass
the embedded agents hold long-lived static credentials in code	pass

Proof card

P11a – NEGATIVE CONTROL: GOVERNANCE INSIDE EVERY AGENT	
Change	production restarts need approval
Edit	incident_agent.py · sha256 aae655907550... → 3a424afcaad3...
Running process	after the edit → restart EXECUTED ← policy not in effect
Redeployed process	restart → PENDING_APPROVAL
All four changes	7 file edits · 17 lines · 7 agent redeloys
Credentials	8 static credential literals in agent code

- ✓ before the edit the embedded agent restarts production
- ✓ BROKEN: after the edit, the running process still restarts production (no central change is possible)
- ✓ only a redeployed process (new code hash) holds the restart
- ✓ the embedded agents hold long-lived static credentials in code

PROOF P11a: NEGATIVE CONTROL · property broken by design

Test assertions: 4/4 passed

P11-C-control-plane

NEGATIVE CONTROL · governance embedded in agents: the property is broken by design

Question. Without a control plane, what does the same set of governance changes cost, and when do they take effect?

Expected. each change takes effect at the next run with no code change

Observed. 4 versions; 0 edits; 0 redeloys

Outcome. HELD

Step ledger (30 rows)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision
1 · t0	control plane		v1	change.published seed	PUBLISHED
2 · t10	control plane		v2	change.published migrate-default- model	PUBLISHED
3 · t12	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.requested	
4 · t12	runtime rt-a · rt- a/pid-1	req bcc78da5ba13	v2	config.applied	
5 · t12	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance-agent.status	run.decision	ALLOW · AGENT_ACTIVE
6 · t13	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance- agent.tools.read_ledger	tool.executed · read_ledger	ALLOW · POLICY_ALLOW
7 · t14	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · agents.finance- agent.tools.flag_transaction	action.executed · flag_transaction	ALLOW · POLICY_ALLOW
8 · t15	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13	v2 · models.profiles.standard.default	model.called · large-model	ALLOW · MODEL_RESOLVED
9 · t15	runtime rt-a · rt- a/pid-1	finance-agent · req bcc78da5ba13		run.finished	
10 · t20	control plane		v3	change.published restart-requires- approval	PUBLISHED
11 · t22	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested	
12 · t22	runtime rt-a · rt- a/pid-1	req cb0531217710	v3	config.applied	
13 · t22	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v3 · agents.incident-agent.status	run.decision	ALLOW · AGENT_ACTIVE
14 · t23	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v3 · agents.incident- agent.tools.query_logs	tool.executed · query_logs	ALLOW · POLICY_ALLOW

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision
15 · t24	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v3 · agents.incident- agent.tools.query_metrics	tool.executed · query_metrics	ALLOW · POLICY_ALLOW
16 · t25	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v3 · models.profiles.standard.default	model.called · large-model	ALLOW · MODEL_RESOLVED
17 · t26	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v3 · agents.incident- agent.tools.restart_service[1]	approval.requested · restart_service	APPROVAL_REQUIRED · POLICY_APPROVAL_REQUIRED
18 · t26	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.finished	
19 · t30	control plane		v4	change.published revoke- observability-mcp	PUBLISHED
20 · t32	runtime rt-a · rt- a/pid-1	support-agent · req afd77e0ac798		run.requested	
21 · t32	runtime rt-a · rt- a/pid-1	req afd77e0ac798	v4	config.applied	
22 · t32	runtime rt-a · rt- a/pid-1	support-agent · req afd77e0ac798	v4 · agents.support-agent.status	run.decision	ALLOW · AGENT_ACTIVE
23 · t33	runtime rt-a · rt- a/pid-1	support-agent · req afd77e0ac798	v4 · agents.support- agent.tools.read_case	tool.executed · read_case	ALLOW · POLICY_ALLOW
24 · t34	runtime rt-a · rt- a/pid-1	support-agent · req afd77e0ac798	v4 · mcp_servers.observability- mcp.status	action.denied · query_logs	DENY · MCP_SERVER_DISABLED
25 · t35	runtime rt-a · rt- a/pid-1	support-agent · req afd77e0ac798	v4 · models.profiles.standard.confidential	model.called · private-model	ALLOW · MODEL_RESOLVED
26 · t35	runtime rt-a · rt- a/pid-1	support-agent · req afd77e0ac798		run.finished	
27 · t40	control plane		v5	change.published suspend-incident- agent	PUBLISHED

Step tick	Source process	Agent request	Version rule	Event	Decision
28 t42	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710		run.requested	
29 t42	runtime rt-a · rt- a/pid-1	req cb0531217710	v5	config.applied	
30 t42	runtime rt-a · rt- a/pid-1	incident-agent · req cb0531217710	v5 · agents.incident-agent.status	run.decision	DENY · AGENT_SUSPENDED

Measures

Measure	Value
files_edited	0
lines_changed	0
redeploys	0
bundle_versions	4
static_credential_literals	0

Assertions (5/5)

Assertion	Result
default model moved at the next run (large-model)	pass
restart held for approval at the next run	pass
observability tools denied at the next run	pass
incident-agent denied at the next run	pass
0 agent files edited, 0 processes restarted, 4 bundle versions	pass

Proof card

P11b – THE SAME FOUR CHANGES THROUGH THE CONTROL PLANE	
Changes	model default · restart approval · revoke MCP · suspend
Published	v2 ... v5 · 4 signed bundle versions
Agent code	0 files edited · 0 redeploys · one runtime process throughout
Effect	each change applied at the next enforcement point

- ✓ default model moved at the next run (large-model)
- ✓ restart held for approval at the next run
- ✓ observability tools denied at the next run
- ✓ incident-agent denied at the next run
- ✓ 0 agent files edited, 0 processes restarted, 4 bundle versions

PROOF P11b: PASS
Test assertions: 5/5 passed

P12-C-governing-the-control-plane

GOVERN THE GOVERNOR · *the control plane's own changes are authorized, validated and on a tamper-evident log*

Question. Who may change the control plane, and is every change, accepted or rejected, on the record?

Expected. illegitimate changes rejected and logged; legitimate ones versioned and signed

Observed. 3 accepted, 7 rejected, all 11 on a verifying chain

Outcome. HELD

Step ledger (11 rows)

Step · tick	Source · process	Agent · request	Version · rule	Event	Decision	Approval · cost	System of record	Audit
1 · t0	control plane		v1	change.published seed	PUBLISHED			82d854efd2e3
2 · t10	control plane			change.rejected self-grant	REJECTED · incident-agent is not a control-plane administrator			0db9c1f11d0d
3 · t11	control plane			change.rejected cross-team	REJECTED · support.lead has no scope over incident-agent			aed0abf8a8ed
4 · t12	control plane		v2	change.published own-scope	PUBLISHED			d0469f2c0541
5 · t13	control plane			change.rejected widen-alone	REJECTED · a widening change needs a second approver			fa0cdfc812a1
6 · t14	control plane			change.rejected widen-self-approved	REJECTED · the second approver must be a different person			0ecebcee2104
7 · t15	control plane			change.rejected glass-widen	REJECTED · a break-glass change may only restrict behaviour			ff7b6fae9f60
8 · t16	control plane			change.rejected plaintext-cred	REJECTED · invalid desired state: tool restart_service: credential must be a secret:// reference, never a value			0698d074c6b4
9 · t17	control plane		v3	change.published suspend-incident-agent	PUBLISHED			bc87d04386c
10 · t18	control plane			change.rejected restore-incident-agent	REJECTED · a widening change needs a second approver			e412d4eb18d4
11 · t19	control plane		v4	change.published restore-incident-agent	PUBLISHED			0488748c3181

Measures

Measure	Value
attempts	10
accepted	3
rejected	7
changelog_rows	11
versions	4
agent_self_grant	rejected
cross_team_change	rejected
own_scope_restriction	accepted
widening_alone	rejected
self_approval	rejected
break_glass_widening	rejected
plaintext_credential	rejected
break_glass_suspension	accepted
restore_with_second_person	accepted
changelog_verifies	yes
tampered_changelog_verifies	no

Assertions (13/13)

Assertion	Result
an agent cannot change the control plane (not an administrator)	pass
scope: a team lead cannot change another team's agent	pass
scope: a team lead may restrict its own agent alone	pass
two-person rule: a widening change needs a second approver	pass
two-person rule: the second approver must be a different person	pass
break-glass may only restrict	pass
validation: a plaintext credential never becomes a version	pass
break-glass suspension accepted from one on-call person	pass
restoring (widening) needs a second person	pass
every attempt is on the change log, accepted or rejected	pass
the change log is hash-chained and verifies	pass
editing one change-log row breaks the chain	pass
every version is signed; no bundle contains a plaintext credential	pass

Proof card

P12 — GOVERNING THE CONTROL PLANE ITSELF

incident-agent	grant itself delete_resource → REJECTED · incident-agent is not a control-plane administrator
support.lead	raise incident-agent's tool quota → REJECTED · support.lead has no scope over incident-agent
support.lead	lower support-agent's tool quota (own scope) → ACCEPTED · v2
platform.admin	widen incident-agent's tools alone → REJECTED · a widening change needs a second approver
platform.admin	platform.admin as its own second approver → REJECTED · the second approver must be a different person
oncall.ic	raise a quota via break-glass → REJECTED · a break-glass change may only restrict behaviour
platform.admin	a plaintext credential in the registry (with sre.lead) → REJECTED · tool restart_service: credential must be a secret:// reference, never a value
oncall.ic	suspend incident-agent (break-glass) → ACCEPTED · v3
oncall.ic	restore incident-agent alone → REJECTED · a widening change needs a second approver
oncall.ic	restore incident-agent with sre.lead → ACCEPTED · v4
Change log	11 rows, hash-chained · verifies; one edited row → chain broken · tamper-evident local log under POC assumptions

- ✓ an agent cannot change the control plane (not an administrator)
- ✓ scope: a team lead cannot change another team's agent
- ✓ scope: a team lead may restrict its own agent alone
- ✓ two-person rule: a widening change needs a second approver
- ✓ two-person rule: the second approver must be a different person
- ✓ break-glass may only restrict
- ✓ validation: a plaintext credential never becomes a version
- ✓ break-glass suspension accepted from one on-call person
- ✓ restoring (widening) needs a second person
- ✓ every attempt is on the change log, accepted or rejected
- ✓ the change log is hash-chained and verifies
- ✓ editing one change-log row breaks the chain
- ✓ every version is signed; no bundle contains a plaintext credential

PROOF P12: PASS

Test assertions: 13/13 passed

Reproduce

```
make verify      # rerun every proof from source; identical to this run (process ids masked)
make evidence    # PROOF VERIFICATION of the published run
make verify-all  # the final gate
make console     # the Lab Console
```

Evidence

Run report

AI Control Plane

PRODUCTION AI ENGINEERING · THE SERIES SO FAR

FOUNDATION

F1 · MCP Tool Sprawl

Your AI agent has 500 MCP tools. Now what?

FOUNDATION

F2 · Layered Architecture

Your agent works in a demo. Why does it break in production?

FOUNDATION

F3 · Headless AI

Your AI shouldn't live inside the UI

TRUST

T1 · Agent Identity

Who is acting, and on whose authority?

TRUST

T2 · Authorization & Policy

What is this agent actually allowed to do?

PREVIOUS

T3 · Human-in-the-Loop

The Human-in-the-Loop chapter

CURRENT

T4 · AI Control Plane

Who governs every agent, and how does one change reach all of them?

NEXT

T5 · Observability & Governance

Your AI agent did something in production. Can you explain exactly what happened?

The full learning map (Start Here)

T4 · Production AI Engineering

Production AI Engineering · T4 · Run report. Every measured number here is substituted at build time from `control_plane_poc/runs/2026-10-03-recorded/facts.json`; the enterprise systems are simulated, every runtime is a real long-lived process, and the run is deterministic. Figures carry their provenance.

- Medium edition Technical deep dive Evidence Check POC README Proof cards
- Run summary Lab Console QA report